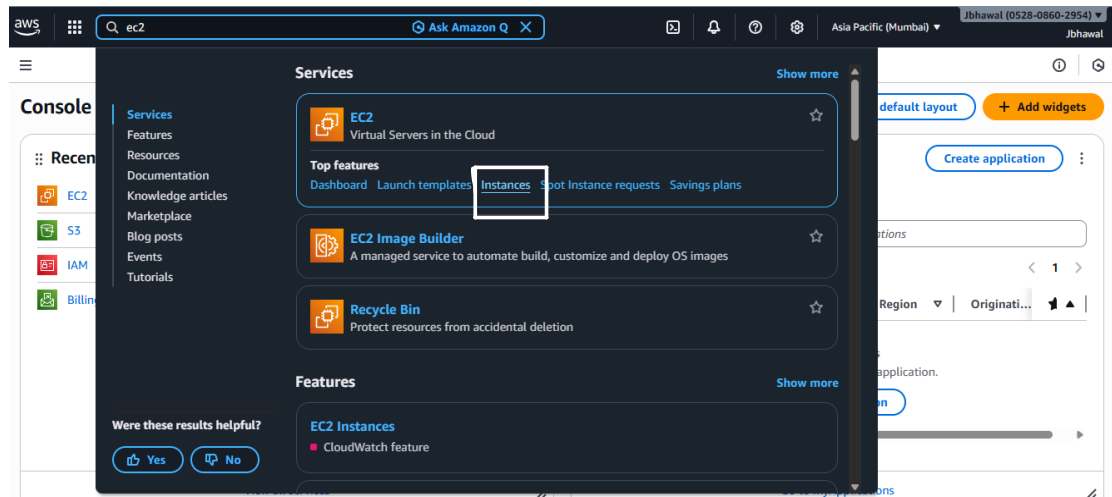
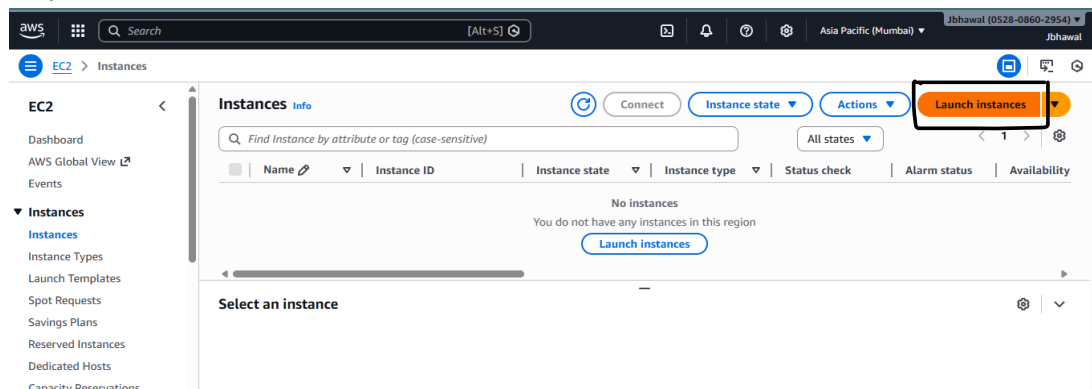


Assignment 9: Deploy a project from Github to EC2.

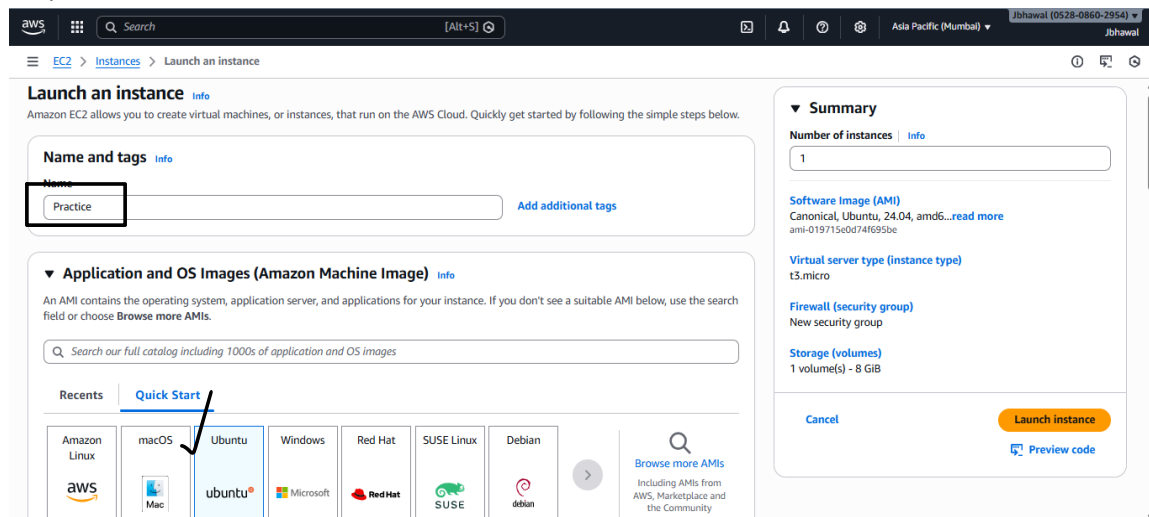
Step 1: Search ec2 and click on Instances under the Services appearing.



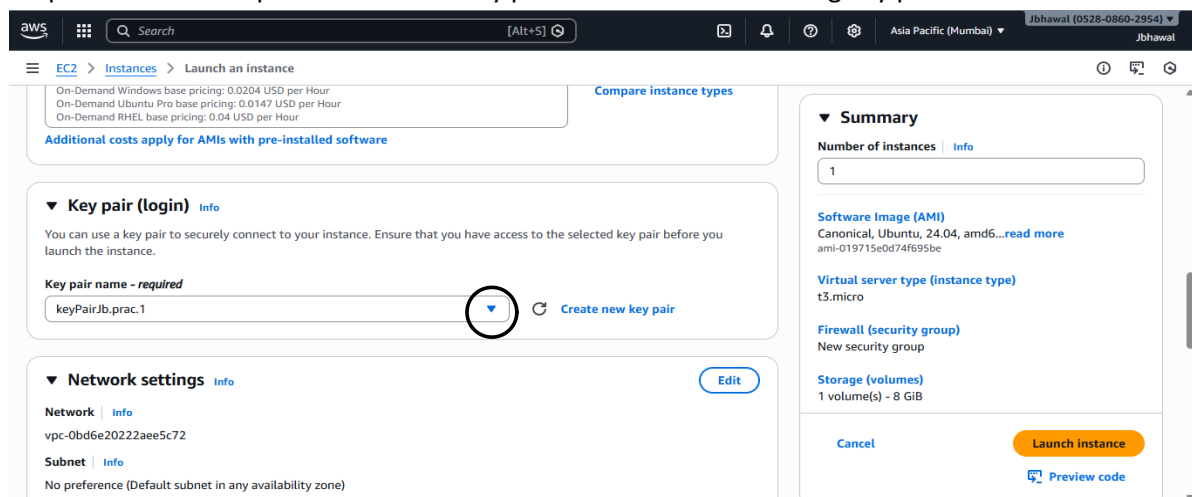
Step 2: Click on Launch instances



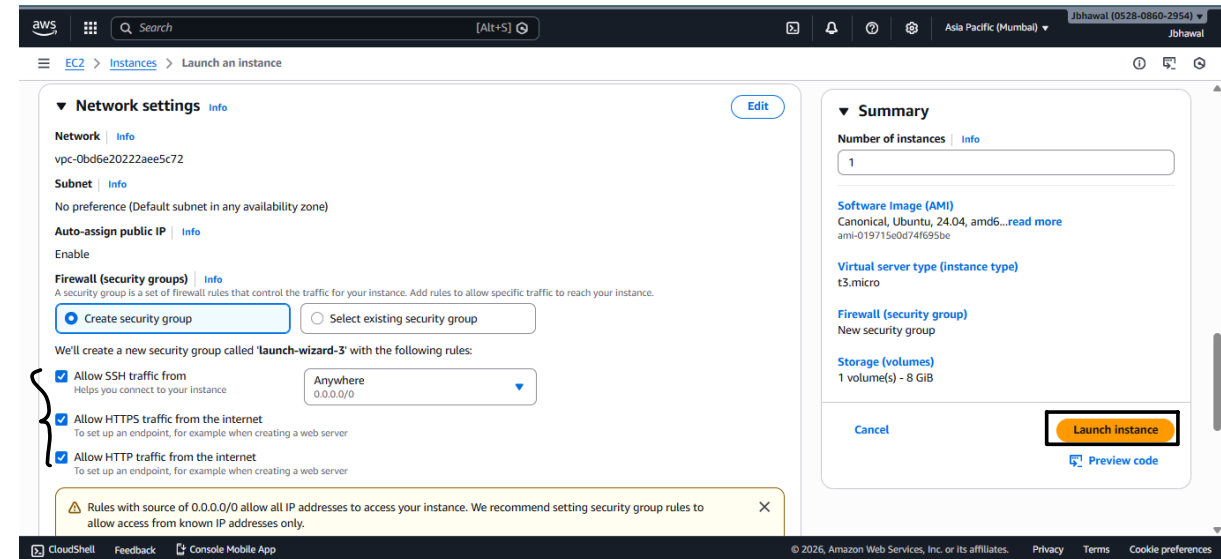
Step 3: Give an instance name and select Ubuntu as the AMI.



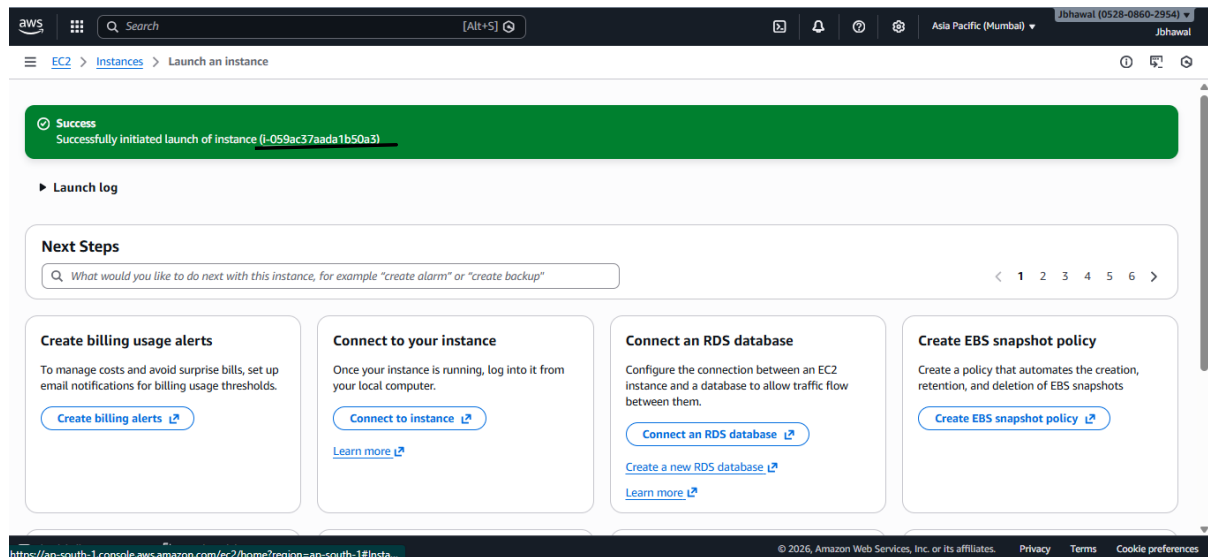
Step 4: From the drop-down menu of Key pair name select an existing key pair or create a new one.



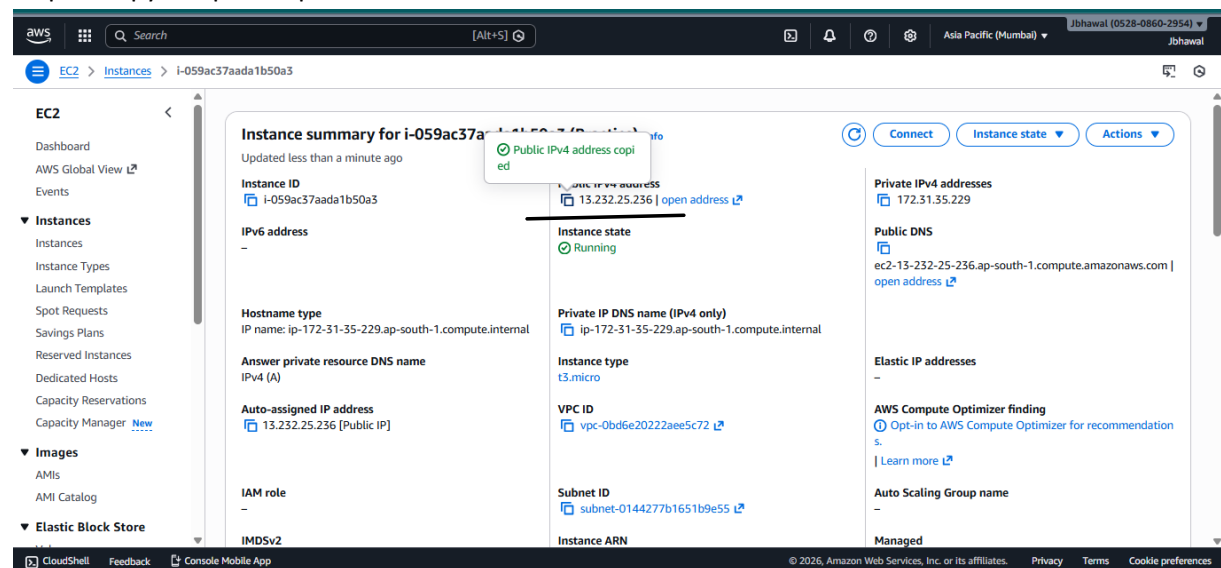
Step 5: Select all the checkboxes under the Security permissions. Then click on Launch instance.



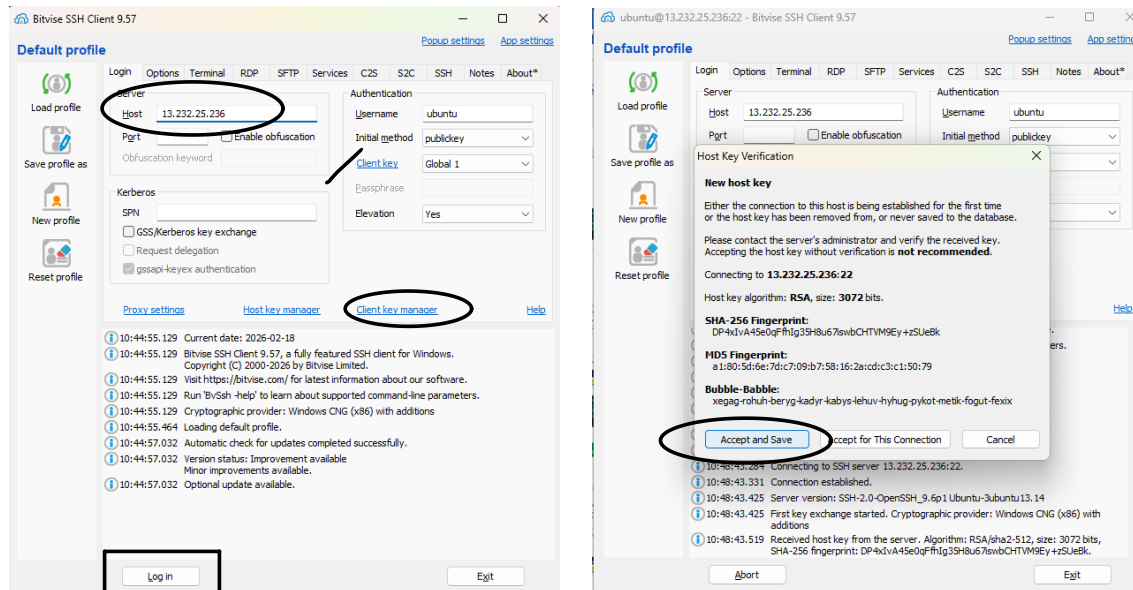
Step 6 : The green banner indicates confirmation of the instance creation. Click on the instance id.



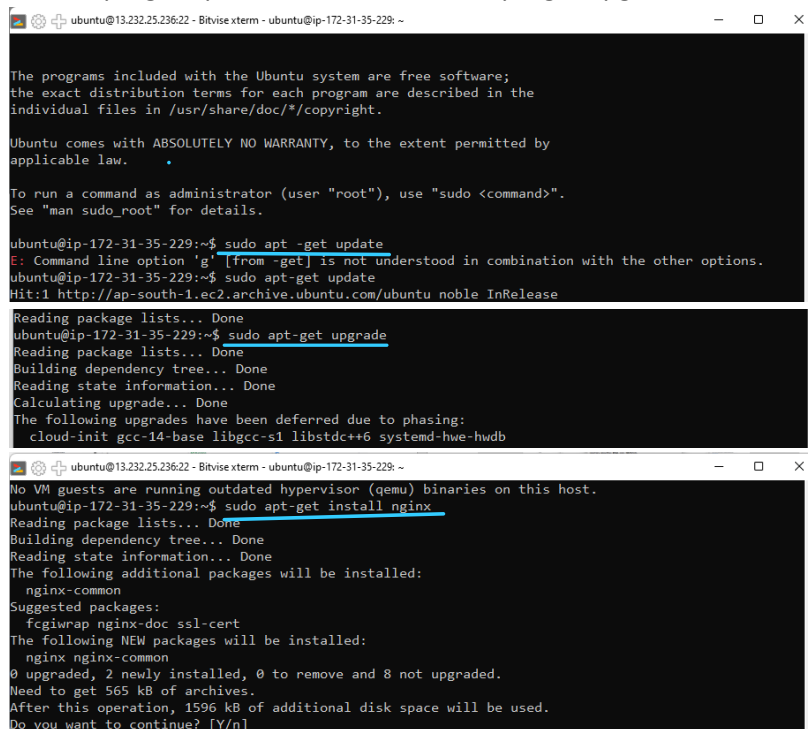
Step 7: Copy the public ipv4 address.



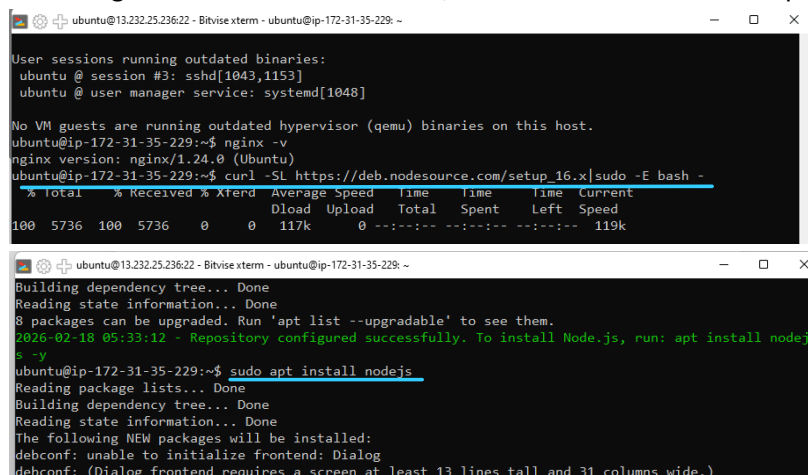
Step 8: Open Bitvise SSH Client and in Host, give the ipv4 address just copied. Select proper Client key. Manage it in the Client key manager. Then click on Login and then click Accept and Save when prompted.



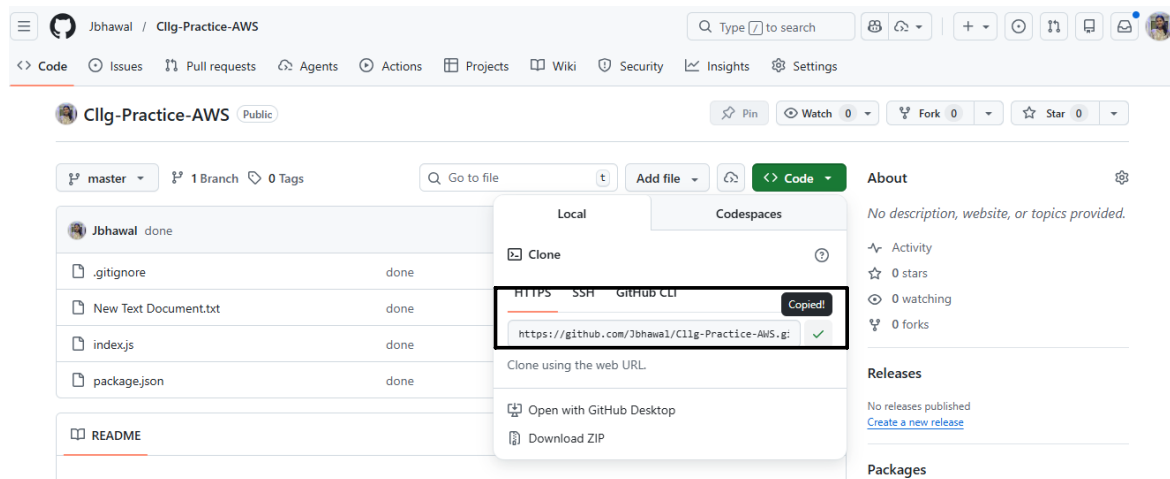
Step 9 : Open the terminal and write the commands-
<sudo apt -get update> and then <sudo apt -get upgrade> and then <sudo apt -get install nginx> Enter Y if prompted.



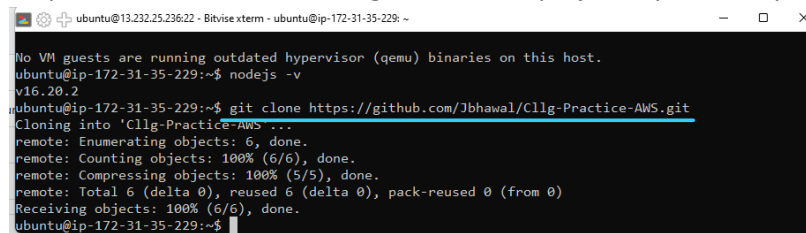
Step 10: To download the setup file of node js write the following command next: curl -SL https://deb.nodesource.com/setup_16.x|sudo -E bash -
Following this command's execution, write the command: sudo apt install nodejs



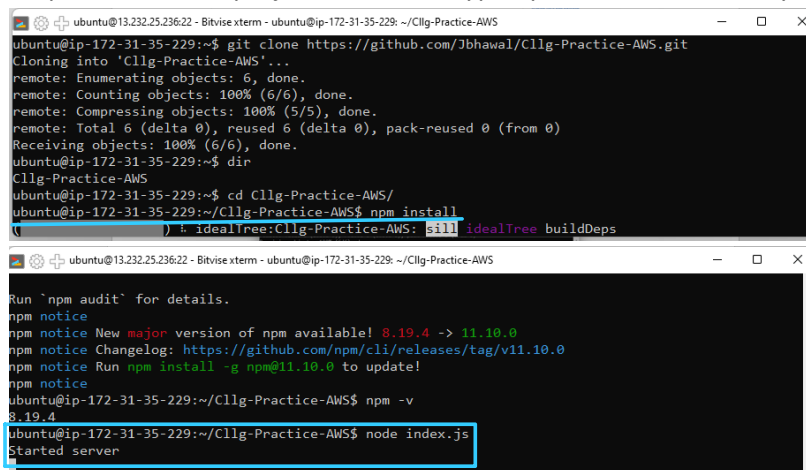
Step 11: Copy the repository URL from GitHub.



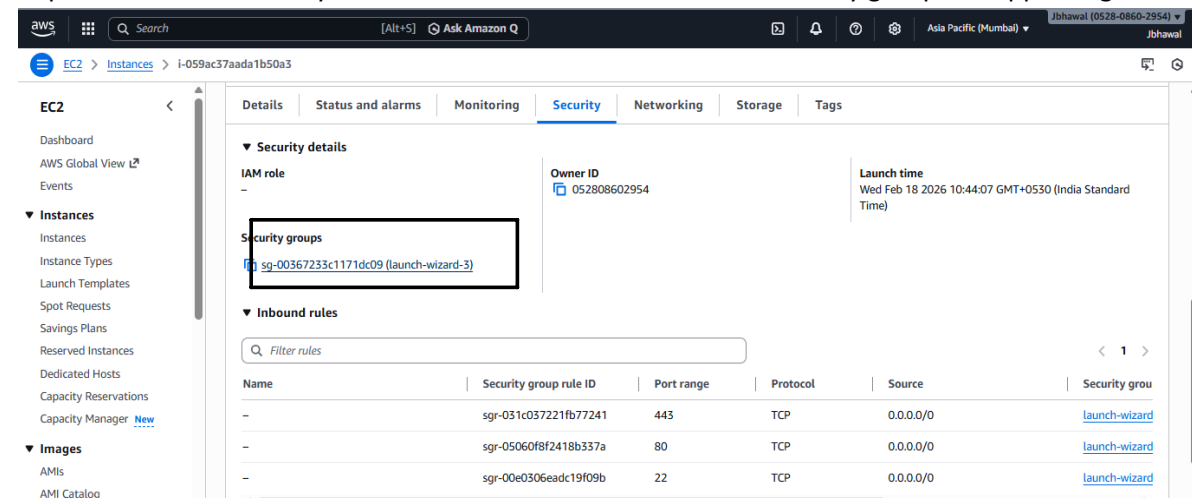
Step 12: Write the command: `git clone <the project repo URL copied>`



Step 13: Go to the project folder and type `npm install` followed by `node index.js`. This starts the server.



Step 14: Go to the Security section of the instance. Click on the Security group link appearing.



Step 15: Scroll down to find Inbound rules section and click on Edit inbound rules.

The screenshot shows the AWS Management Console for a security group named 'sg-00367233c1171dc09 - launch-wizard-3'. The 'Inbound rules' tab is active, displaying a table of existing rules:

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-031c037221fb77241	IPv4	HTTPS	TCP	443
-	sgr-05060f8f2418b337a	IPv4	HTTP	TCP	80
-	sgr-00e0306eadc19f09b	IPv4	SSH	TCP	22

The 'Edit inbound rules' button is highlighted with a red box.

Step 16: Click on Add rule

The screenshot shows the 'Edit inbound rules' page. The 'Add rule' button is highlighted with a red box. Below the table of rules, there is a warning message: 'Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.'

Step 17: Enter 4000 in Port range and select 0.0.0.0/0 as source and then click on Save rules.

The screenshot shows the 'Edit inbound rules' page with a new rule being added. The 'Port range' is set to '4000' and the 'Source' is set to '0.0.0.0/0'. The 'Save rules' button is highlighted with a red box.

Step 18: In a new tab's address bar paste the ipv4 address copied earlier followed by :4000 to see the project running.

The screenshot shows a web browser with the address bar containing the URL '13.232.25.236:4000'. The page displays the nginx welcome message: 'If you see this page, the nginx web server is successfully installed and working. Further configuration is required. For online documentation and support please refer to nginx.org. Commercial support is available at nginx.com. Thank you for using nginx.'